Reg. No. | | | | | | | | | | | [14MAEPR101]

FIRST SEMESTER M.TECH EXAMINATION, FEBRUARY 2016

(AIRCRAFT PROPULSION)

GAS DYNAMICS

Time: 03 Hours **Maximum Marks:100**

Instructions :
- **i. Answer any Five questions selecting at least two questions from each part.**
- **ii. Draw neat sketches wherever necessary**
- **iii. Make suitable assumptions where necessary**

PART – A

1. a. Derive the equations for entropy change per unit Mass of gas during a process. **10**

b. A Supersonic aircraft is flying horizontally at the altitude of 3000m with a constant flight speed of 2000 km /hr. find the time taken to hear the sound waves from aircraft at a ground observation post after the aircraft has passed directly over it. Assume average temperature of atmospheric air below 3000m altitude as 27^0C. **05**

c. Air enters a constant area duct at M_1=3, P_1=1 atm and T_1=300K. How much heat per unit mass must be added to the flow inside the duct to choke the flow? **05**

2. a. Derive the integral form of the momentum equation. **10**

b. Drive the equations for T_0/T and P_0/P and the relation between them. **10**

3. a. A convergent –divergent nozzle is designed to operate with an exit Mach number of 1.75. The nozzle is supplied from an air reservoir at $68x10^5$ N/m^2 (abs). assuming one dimensional flow, calculate : **10**

- **i.** The maximum back pressure to choke the nozzle
- **ii.** The range of back pressure over which a normal shock will appear in the nozzle
- **iii.** The back pressure for the nozzle to be perfectly expanded to design M and
- **iv.** The range of back pressure for supersonic flow at the nozzle exit plane.

b. Derive the area velocity relation **10**

$$\frac{dA}{A}=\left(M^2-1\right)\frac{du}{u}$$

What do you infer from this relation?

4. a. Consider a streamline with M_1=4.0 and P_1=1atm. Calculate the percentage loss in total pressure in the following two different shock structures encountered by the streamline.

- **i.** A single normal shock wave.
- **ii.** An oblique shock with $\beta=40^0$, followed by a normal shock.

b. Air flows through a frictionless adiabatic convergent- divergent nozzle. The air stagnation pressure and temperature are 7×10^5 P_a and 500K. The diverging portion of the nozzle has an area ratio of $\frac{A_e}{A_t}=11.91$. A normal shock wave stands in the divergent portion of the nozzle where the $M_a=3$ Determine the Mach number, static pressure and temperature at the nozzle exit plane. **10**

PART - B

5. a. Explain briefly with neat sketches regular reflection of oblique shock from solid boundary and its representation on a P-θ diagram. **10**

b. An incident shock wave with wave angle 30^0 infringes on a straight wall. If the upstream flow properties are M1=2.8, P_1=1 atm and T_1=300K, calculate the pressure, temperature, Mach number and total pressure downstream of the reflected wave. **10**

6. a. Derive the governing equation for prendtl Meyer expansion. State the assumptions. **10**

b. Consider a supersonic flow with an upstream Mach number of 3 and pressure of 1 atm. The flow is first expanded around an expansion corner with $\theta=20^0$ and then compressed through a corner with equal angle so that it is returned to its original direction. Calculate the Mach number and pressure downstream of the compression corner. **05**

c. Air at a pressure of 50KPa and a temperature of 300K is to be isentropically expanded from a Mach number of 1.5 to achieve a pressure of 20Kpa. Find out the required flow deflection angle, the final Mach number and temperature of the gas.

7. a. Explain the unit processes of method of characteristics for the following cases: **10**

i. Internal flow

ii. Wall point

b. Given that the governing equation for the in viscid isentropic supersonic flow is governed by $\left[1-\frac{u^2}{a^2}\right]\phi_{xx}+\frac{2uv}{a^2}\phi_{xy}+\left[1-\frac{V^2}{a^2}\right]\phi_{yy}=0$

and given that the velocity potential ϕ is a function of x and y, derive the differential form of compatibility equation's. **10**

8. a. Write a short note on methods of flow visualization in high speed compressible flows. **05**

b. A stream of air whose $M_a=2$ is suddenly deflected and decelerated by a plane oblique shock wave originating from the tip of a stationary wedge. A schlieren photo shows that the wave angle is 45^0 to the original forward direction of the flow. Using the normal shock equations, estimate the wedge angle. **10**

c. The reservoir pressure of supersonic wind tunnel is 10 atm. A pitot tube inserted in the test section measures a pressure of 0.627 atm. Calculate the test section Mach number and area ratio. **05**